<u>Linux File Permissions Management</u>

1. Objective

Review and modify file and directory permissions in a Linux environment to align with organizational security requirements and the principle of least privilege.

2. Taks performed

A) *Reviewed existing permissions.*

- Used „ls -la” to inspect files, directories, hidden files, ownership and permissions settings

B) Modified file permissions

- Removed unnecessary write permissions from other users. Used „chmod o-w project_k.txt”
- Verified the changes using „ls -la”

C) Secured a hidden file

- Used „chmod u-w, g-w, g+r .project_x.txt” to modified permissions on a hidden file. The goal was to remove write access while preserving appropriate read permissions

D) Restricted directory access

- removed execute permissions for the group on the drafts directory using „chmod g-x drafts” - this ensured that only authorized users could access directory contents.

3. Security Concepts demonstrated

- Linux file permissions
- User, group and other access controls
- Principle of least privilege
- File system hardening
- Access management
- Security verification using command-line tools

4. Outcome

Successfully reviewed and modified file and directory permissions to meet organizational security requirements and reduce unauthorized access risks.